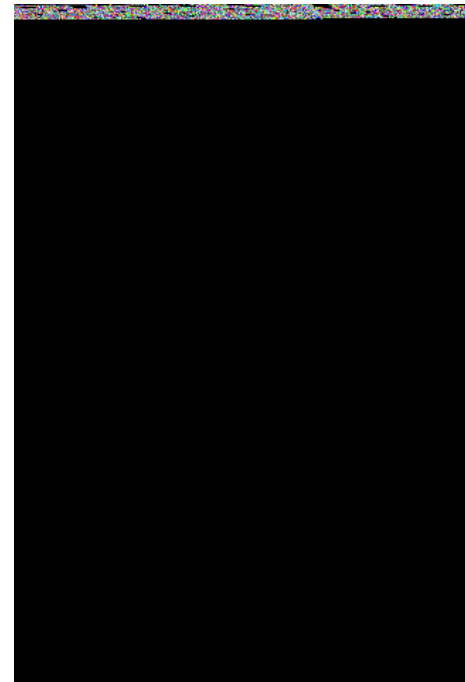


# Pilot operated pressure relief valve

What is pilot operated pressure relief valve ?



## Two Separate Valves

- Main Valve : For Flow.
- Pilot Valve: For Pressure

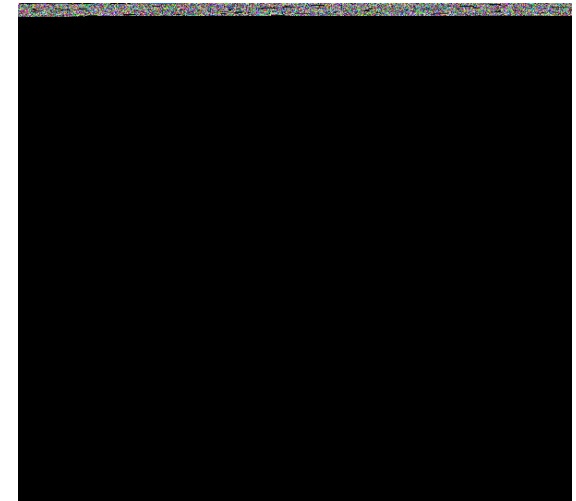
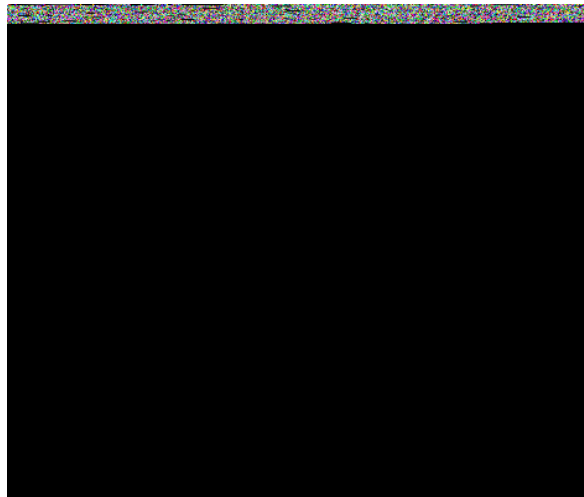
# Pilot operated pressure relief valve

What is pilot operated pressure relief valve ?

The pressure is supplied from the upstream side (the system being protected) to the dome often by a small pilot tube. The downstream side is the pipe or open air where the PORV directs its exhaust. The outlet pipe is typically larger than the inlet.

The upstream pressure tries to push the piston open but it is opposed by that same pressure because the pressure is routed around to the dome above the piston. The area of the piston on which fluid force is acting is larger in the dome than it is on the upstream side; the result is a larger force on the dome side than the upstream side. This produces a net sealing force.

The pressure from the pilot tube to the dome is routed through the actual control pilot valve. There are many designs but the control pilot is essentially a conventional PRV with the special job of controlling pressure to the main valve dome. The pressure at which the control pilot relieves at is the functional set pressure of the PORV. When the pilot valve reaches set pressure it opens and releases the pressure from the dome. The piston is then free to open and the main valve exhausts the system fluid. The control pilot opens either to the main valve exhaust pipe or to atmosphere.



# Pilot operated pressure relief valve

What is the advantage of pilot operated pressure relief valve ?

